

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

Machine Learning (ML)

A branch of AI where computers learn from data and past experiences to improve performance over time without being explicitly programmed. Computers identify patterns and make predictions based on examples.

Examples:

- Email spam detection automatically filters unwanted emails.
- Movie recommendations on platforms like Netflix.
- Stock price predictions using historical data.

Deep Learning (DL)

A specialized part of machine learning that uses neural networks inspired by the human brain. It can process large amounts of data to recognize complex patterns and make decisions more accurately.

Examples:

- Self-driving cars recognizing roads and pedestrians.
- Voice assistants like Siri and Google Assistant understanding speech.

Natural Language Processing (NLP)

AI technology that allows computers to understand, interpret, and respond in human language.

Examples:

- Chatbots answering customer queries online.
- Auto-complete suggestions in messaging apps.

- Language translation using Google Translate.

Computer Vision

AI technology that enables computers to analyze, interpret, and understand images and videos. It helps in tasks like object recognition, facial recognition, and visual data analysis.

Examples:

- Facial recognition for unlocking phones.
- Security cameras detecting suspicious activity.
- Image-based search engines like Google Images.

Robotics

The science and technology of designing, building, and programming robots. Robots can perform tasks autonomously or semi-autonomously, and AI can help them make decisions and adapt to new situations.

Examples:

- Cleaning robots for homes.
- Industrial robots assembling cars in factories.
- Delivery drones navigating autonomously to deliver packages.